Time-Series Analysis

EES 4891/5891

Probability & Statistics for Geosciences

Jonathan Gilligan

Class #26: Thursday, April 10 2025

Learning Goals

Learning Goals

- Understand what stochastic time-series are
  - and how they're different from deterministic time-series
- Understand the difference between stationary and non-stationary time-series
- Understand autocorrelation and partial autocorrelation
- Understand the three major types processes that produce stationary time series:
  - Autoregressive (AR)
  - Moving-Average (MA)
  - Mixed Autoregressive Moving-Average (ARMA)
- Understand how to use the Box-Jenkins method to identify AR and MA time series.

Stochastic Time Series

Stochastic Time Series

- Deterministic process: y is determined by some function: $y(t) = f(t)$.
  - y is completely predictable if you know the function f.
- Stochastic process: y is the result of some random process. y is not perfectly predictable, but if we know the random process, we can predict y with some imperfect degree of certainty.
  - We often consider processes that are the sum of a deterministic and a stochastic process.
    - We can subtract the deterministic process and analyze the pure stochastic process.

- Noise: In time-series analysis, noise means a random process with certain spectral characteristice:
  - White noise: If we take the Fourier transform of the random process, the power-spectrum is uniform across the range of frequencies

    White noise: 0:00 / 0:10
  - Red noise, pink noise: The power spectrum is greater at low frequencies and decreases at higher frequencies.

    Pink noise: 0:00 / 0:10

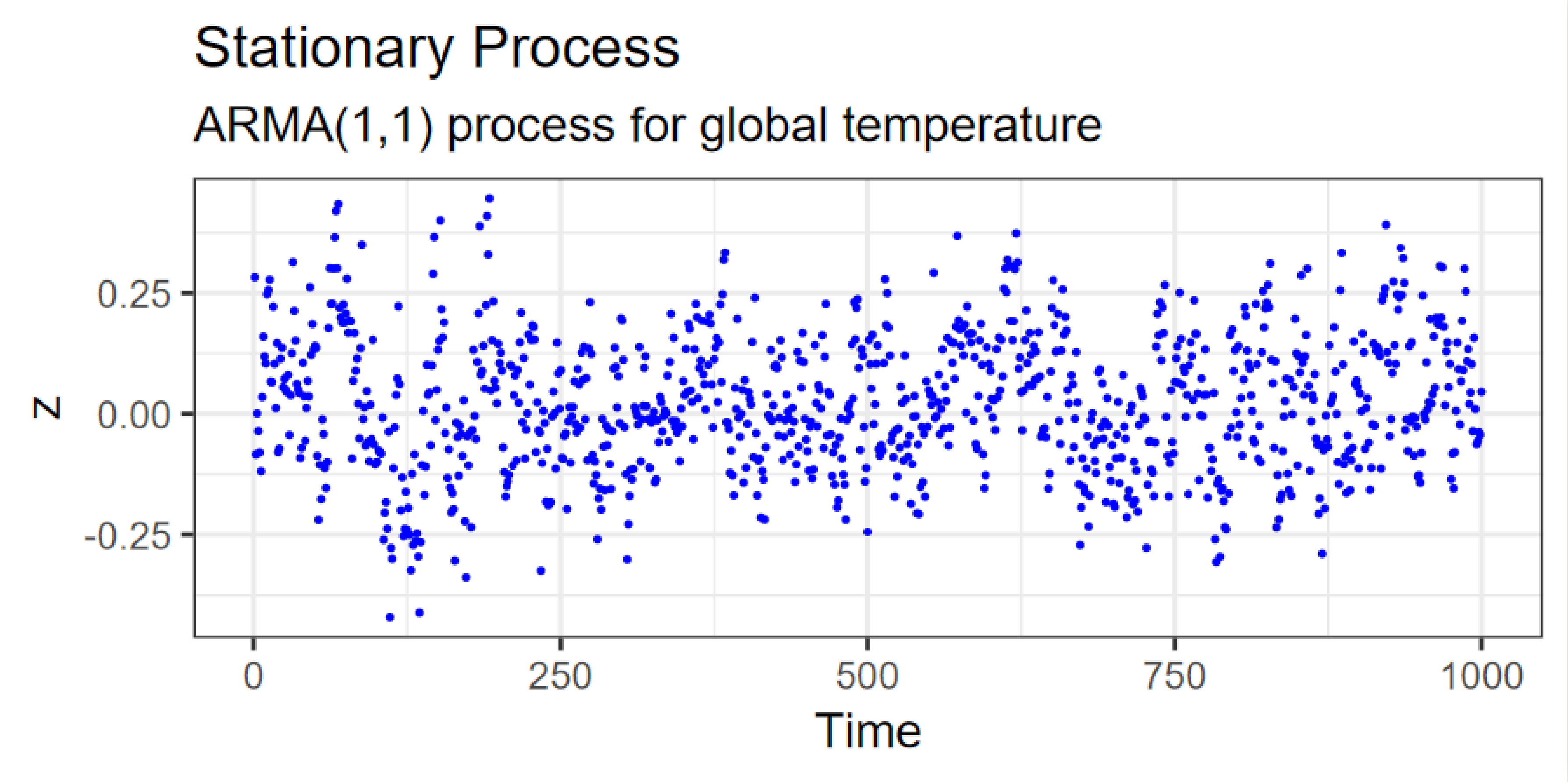
    Red noise: 0:00 / 0:10
    - Red noise is also known as “Brown noise” because it has the same power spectrum as Brownian motion

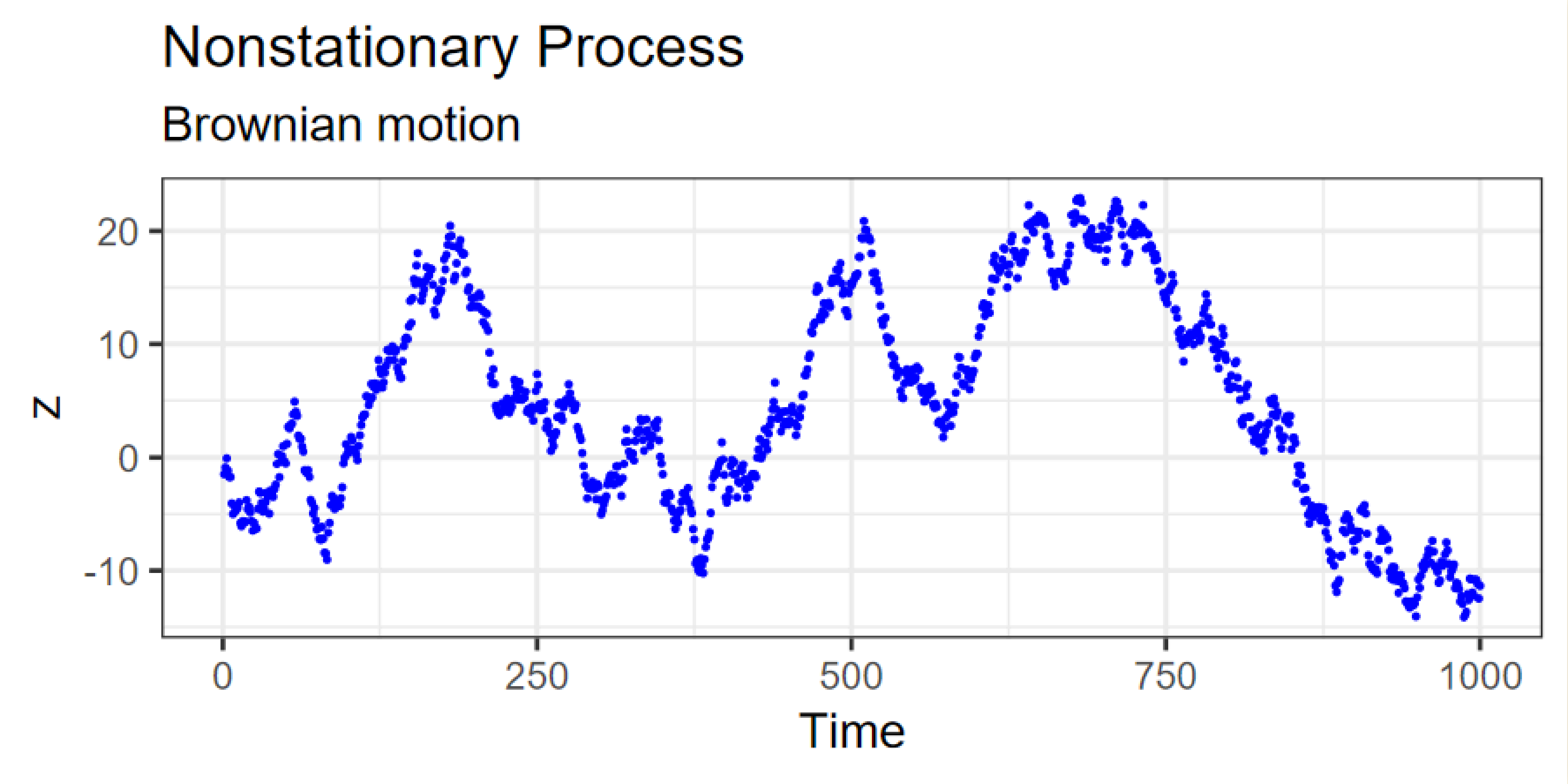
Properties of Time Series Processes

- Pure stochastic process z_t:

$$\mu_t = E(z_t) \qquad \text{mean}$$
$$\sigma_t^2 = \text{Var}(z_t) = E((z_t - \mu_t)^2) \qquad \text{variance}$$

- Stationaray process: μ and σ^2 don't change with time.
- Nonstationary process: μ_t and/or σ_t^2 change with time.





Autocorrelation

Autocorrelation

- Autocorrelation:

$$\rho(t_1, t_2) = \frac{E\left((z_{t_1} - \mu_{t_1})(z_{t_2} - \mu_{t_2})\right)}{\sigma_{t_1}\sigma_{t_2}}$$

  - For a stationary process, this simplifies to

    $$\rho(\tau) = \frac{E\left((z_{t+\tau} - \mu)(z_t - \mu)\right)}{\sigma^2}$$

    where τ is called the “lag”.

  - For discretely sampled z_i, $i = 1, 2, \ldots, N$:

    $$\rho_k = \frac{E\left((z_{i+k} - \mu)(z_i - \mu)\right)}{\sigma^2},$$

    where $k = 1, 2, \ldots, N-1$

Stationary Stochastic Processes

Stationary Stochastic Processes

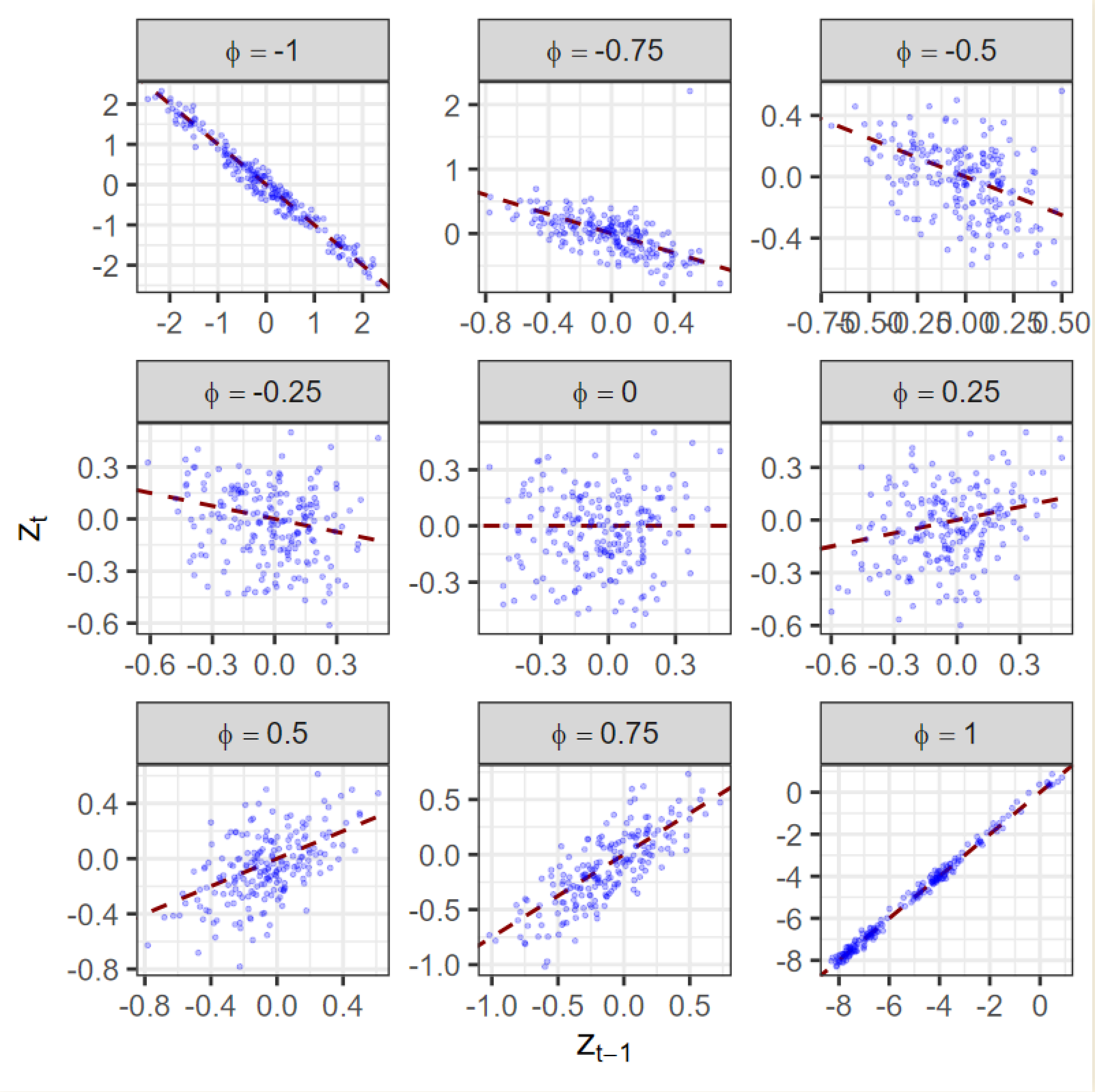
- There are three principal types of models for stationary stochastic processes
  1. Autoregressive (AR) models
  2. Moving-Average (MA) models
  3. Combined Autoregressive Moving-Average (ARMA) models
- We can extend these for nonstationary processes:
  4. Autoregressive Integrated Moving-Average (ARIMA) models
  5. Seasonally-adjusted Autoregressive Integrated Moving-Average (SARIMA) models
- We can use autocorrelation analysis to help identify which kind of model would best represent our data (Box-Jenkins method)
- For this class, we will stick with stationary models: AR, MA, and ARMA

Autoregressive (AR) Processes

- AR(1): Order-1 autoregressive process:

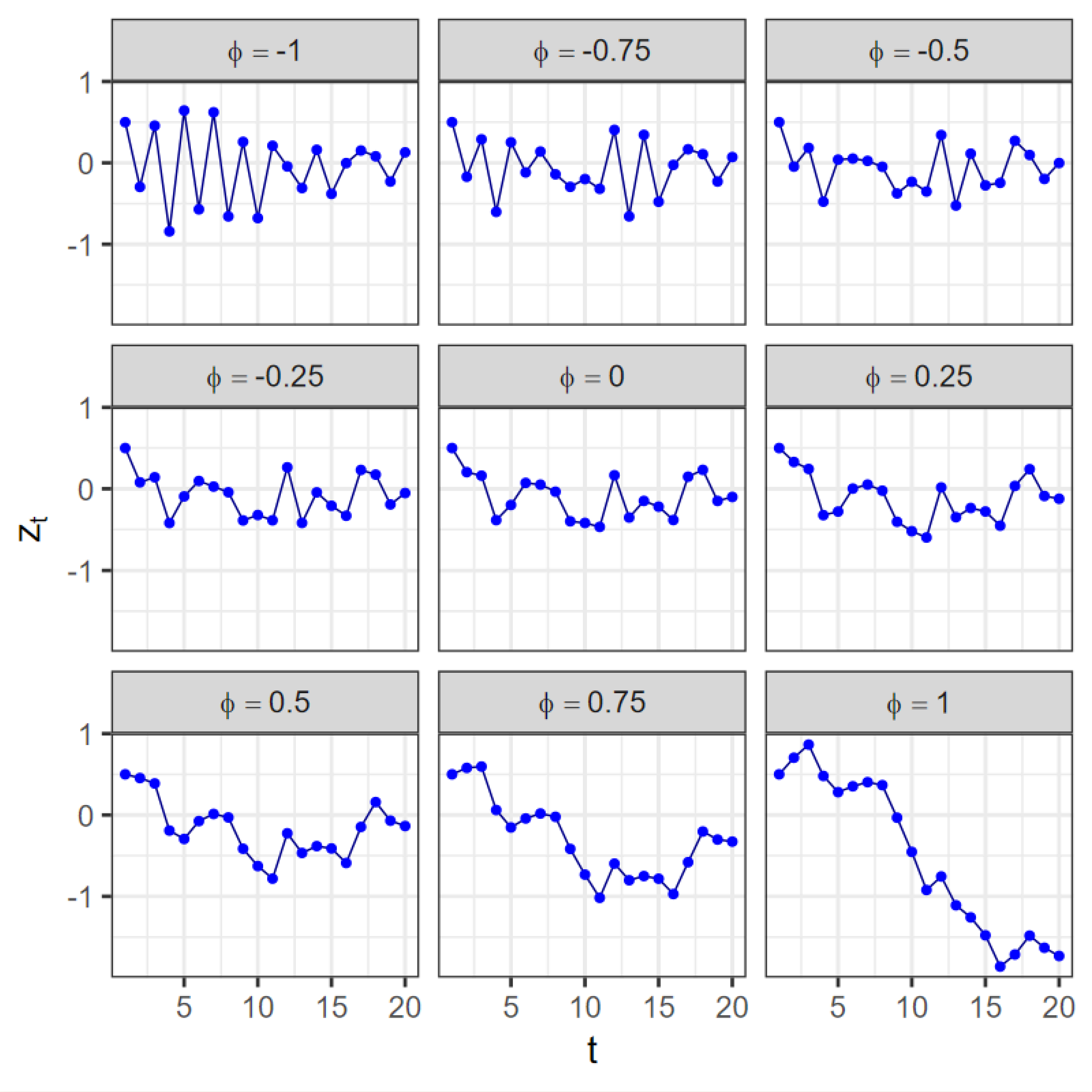
$$z_t = \phi x_{t-1} + \varepsilon_t$$
$$\phi = \rho_1 \quad \text{lag-1 autocorrelation}$$
$$\varepsilon_t \sim \mathcal{N}(0, \sigma)$$

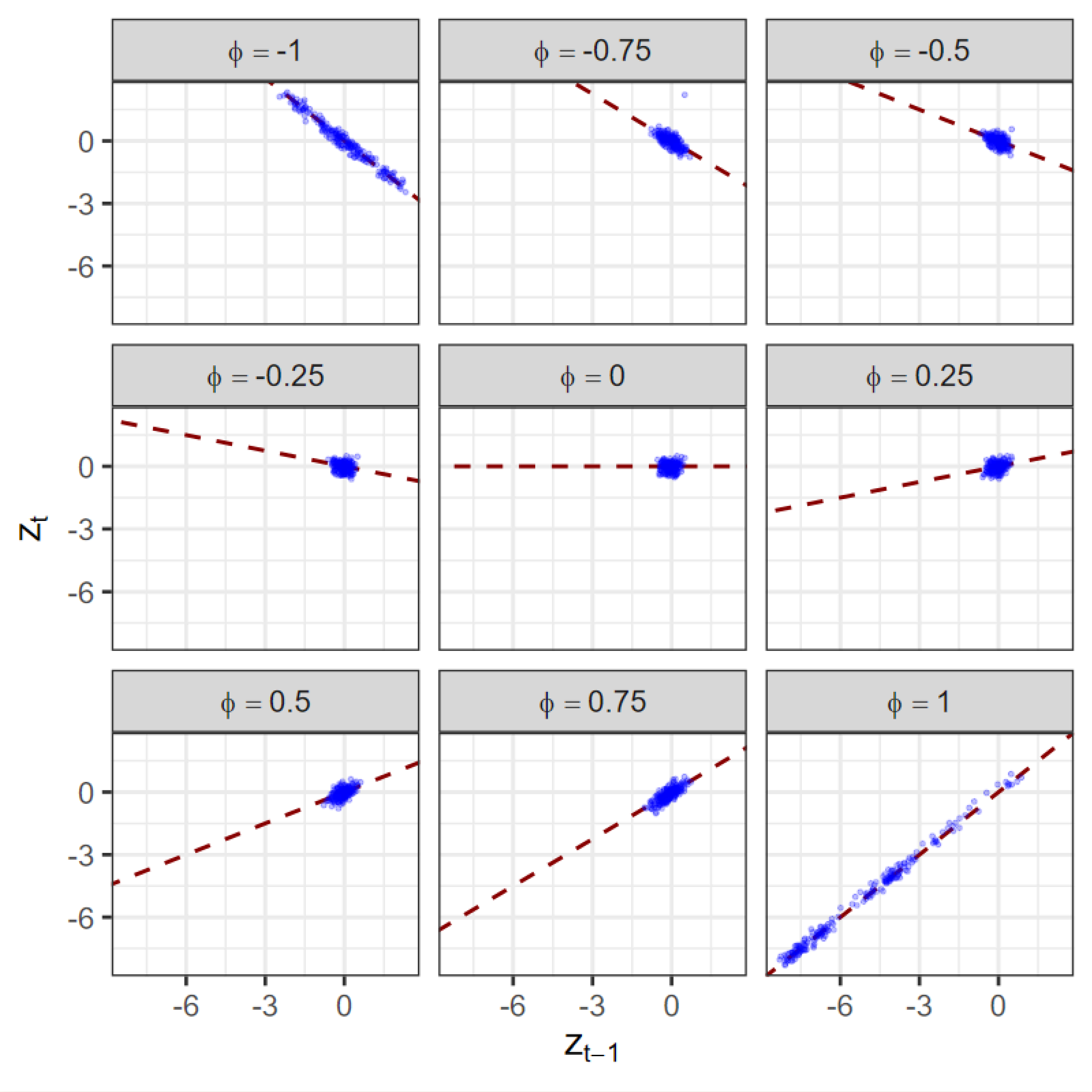
- ϕ represents process "memory".
  - $\phi = 0$: no memory, z_t are independent, normally distributed.
  - Larger $\phi \rightarrow$ smoother.



- Regress z_t against z_{t-1}: slope = ϕ.

AR(1) Processes





- 20 points from the time-series

Higher-Order AR Processes

- AR(p): p-order autoregressive process

$$z_t - \mu = \sum_{i=1}^{p} \phi_i (z_{t-i} - \mu) + \varepsilon_t$$

  - z_t depends only on the previous p observations, and is conditionally independent of anything before $t - p$, given the p values

$$z_{t-p}, z_{t-(p-1)}, \ldots, z_{t-2}, z_{t-1}.$$

- Conditional independence:
  - The previous observations affect z_t, because they led up to it.
  - But there are many different sets of previous z values that could have led up to

$$z_{t-p}, z_{t-(p-1)}, \ldots, z_{t-2}, z_{t-1}.$$

    - Once you specify these p values, z_t does not depend on what specific sequence of previous z values led up to them.
    - *Markov process*: finite memory.
      - Each value only depends on a finite number of previous values.

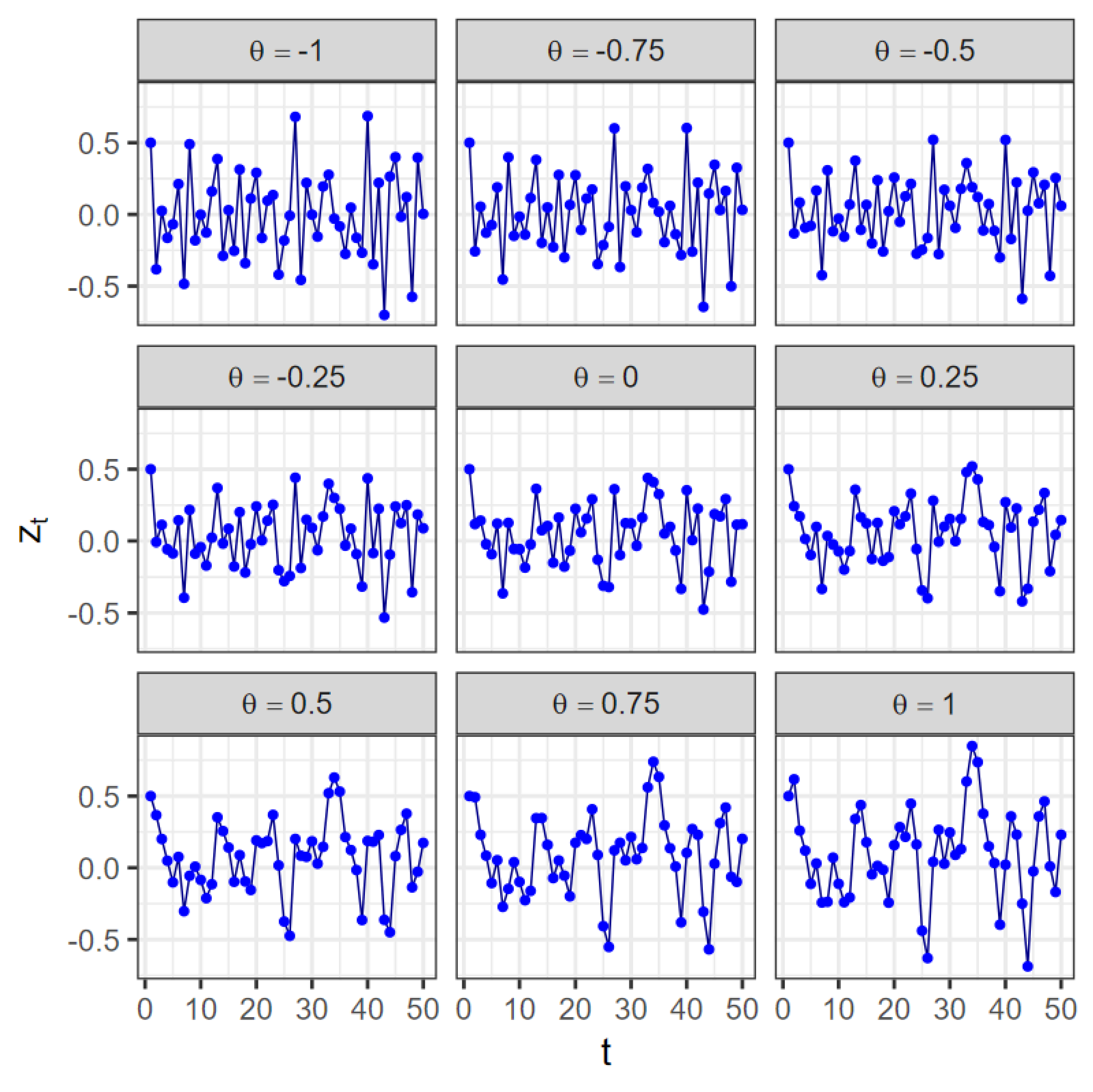
Moving-Average Processes

Moving-Average Processes

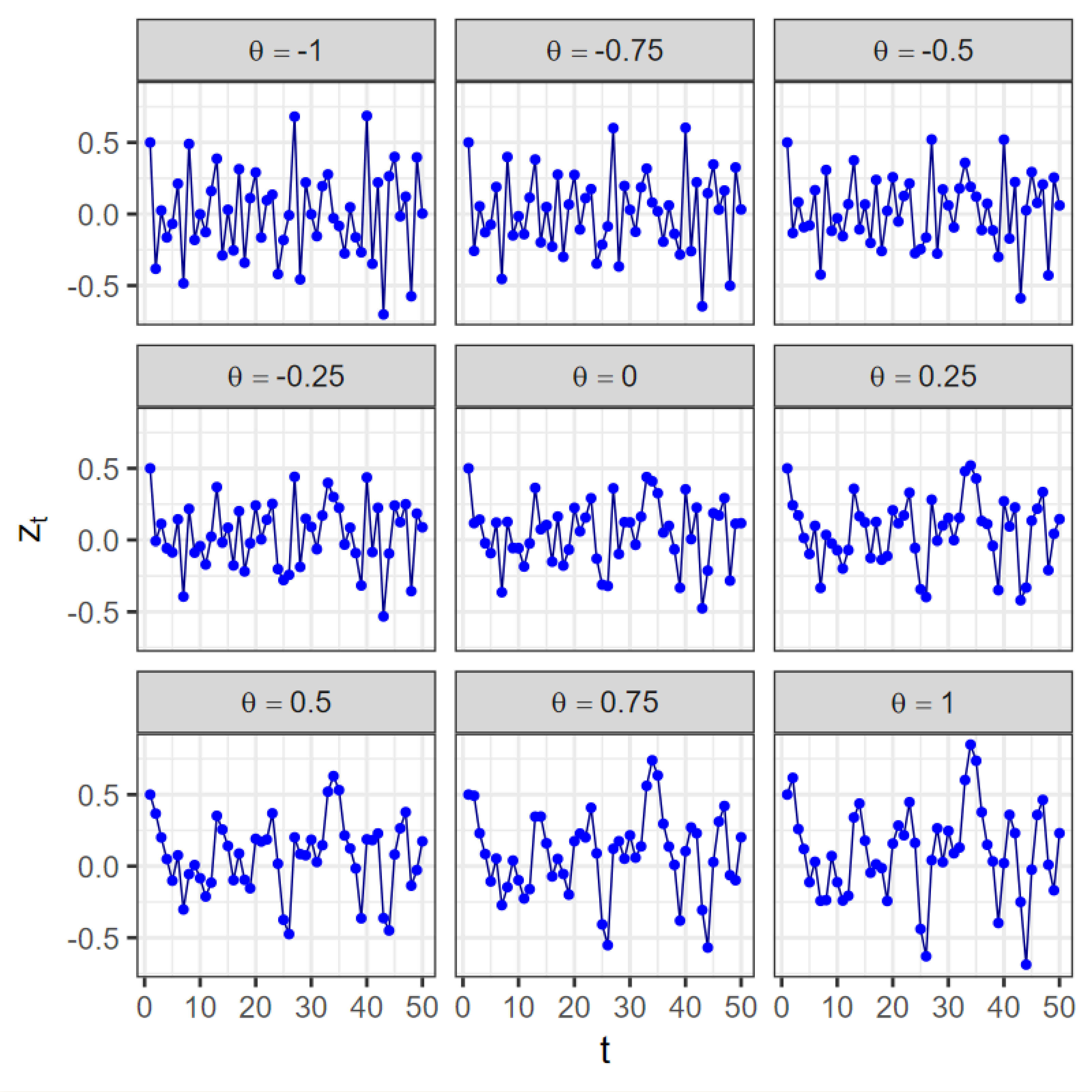
- MA(1): Order-1 moving-average process:

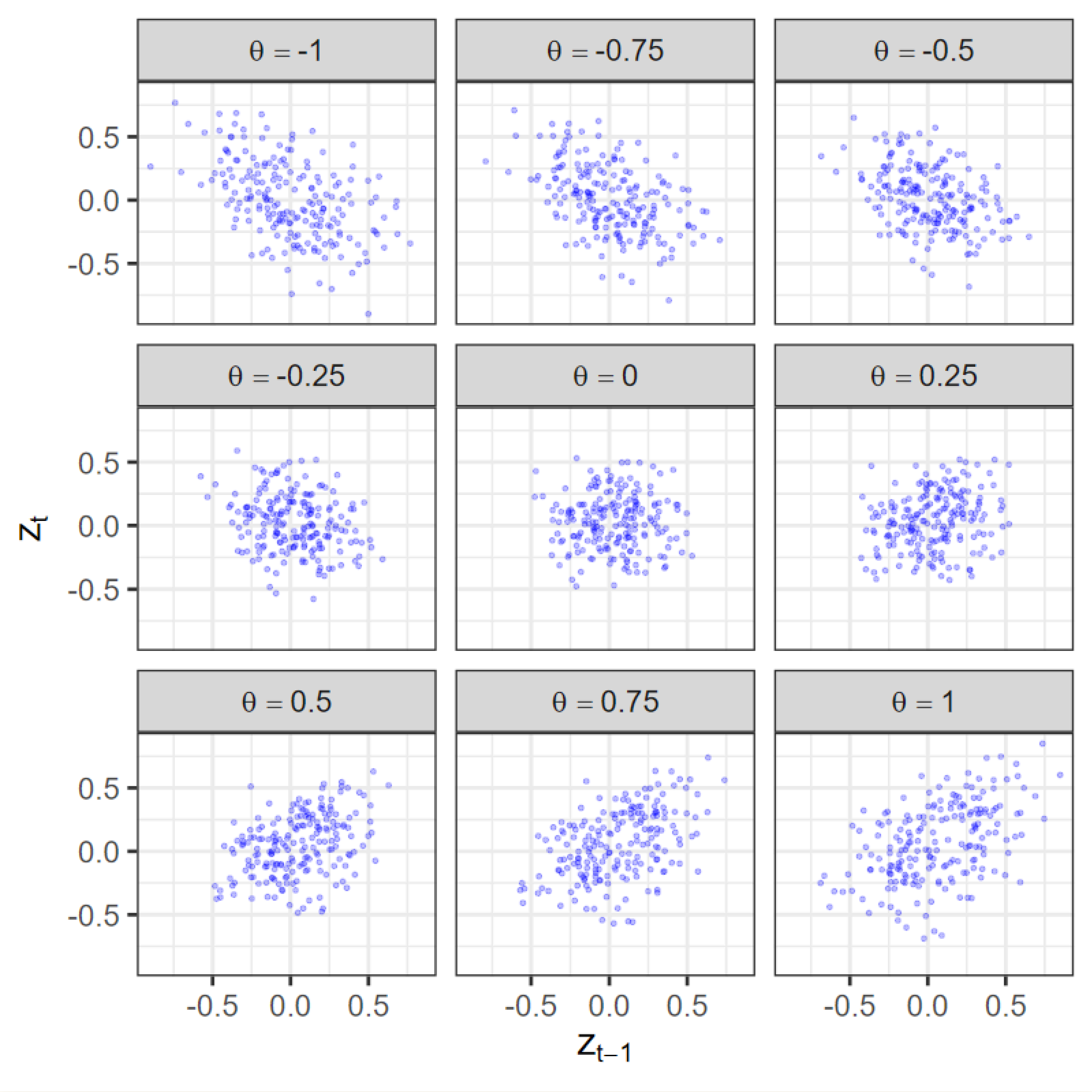
$$z_t = \mu + \theta \varepsilon_{t-1} + \varepsilon_t$$
$$\varepsilon_t \sim \mathcal{N}(0, \sigma)$$

- θ represents a “filter coefficient”.
  - The random part of each value is a weighted average of the previous noise terms plus a new noise term.



MA(1) Processes





- 50 points from the time-series

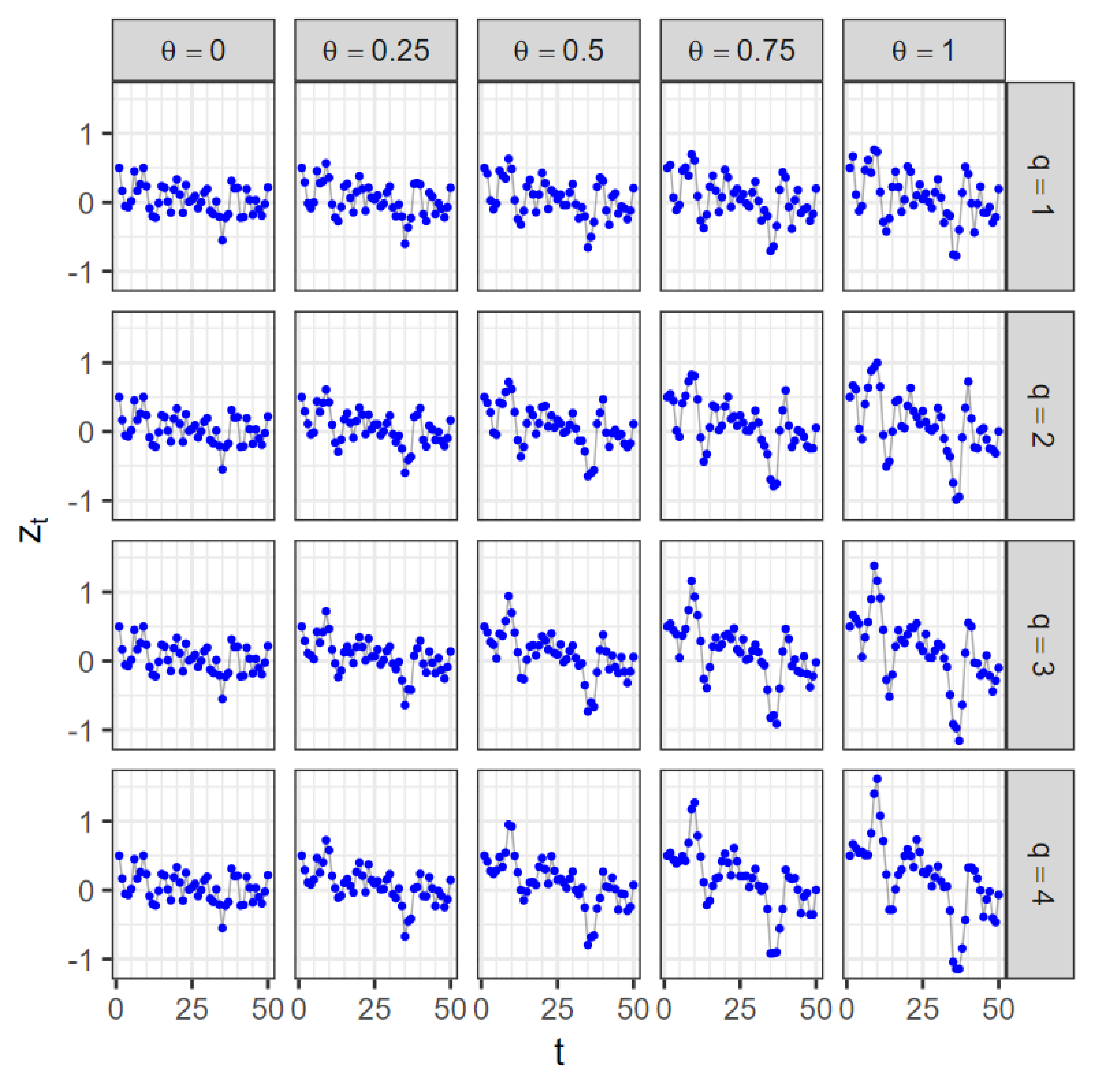
Higher-Order MA Processes

- AR(q): q-order moving-average process

$$z_t - \mu = \sum_{i=1}^{q} \theta_i \varepsilon_{t-i} + \varepsilon_t$$

  - z_t depends on a weighted average of the previous q noise values and is conditionally independent of anything before $t - q$, given the q values

$$\varepsilon_{t-q}, \varepsilon_{t-(q-1)}, \ldots, \varepsilon_{t-2}, \varepsilon_{t-1}.$$



Mixed Autoregressive Moving-Average Processes

Mixed Autoregressive Moving-Average Processes

- ARMA(p, q) process:

$$z_t - \mu = \sum_{i=1}^{p} \phi_i (z_{t-i} - \mu) + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \varepsilon_t$$

Identifying the Process

Setting Up

- Install packages if necessary:

```
install.packages(c("tsibble", "fable", "feasts"))
library(tsibble)
library(fable)
library(feasts)
```

- Download data files:

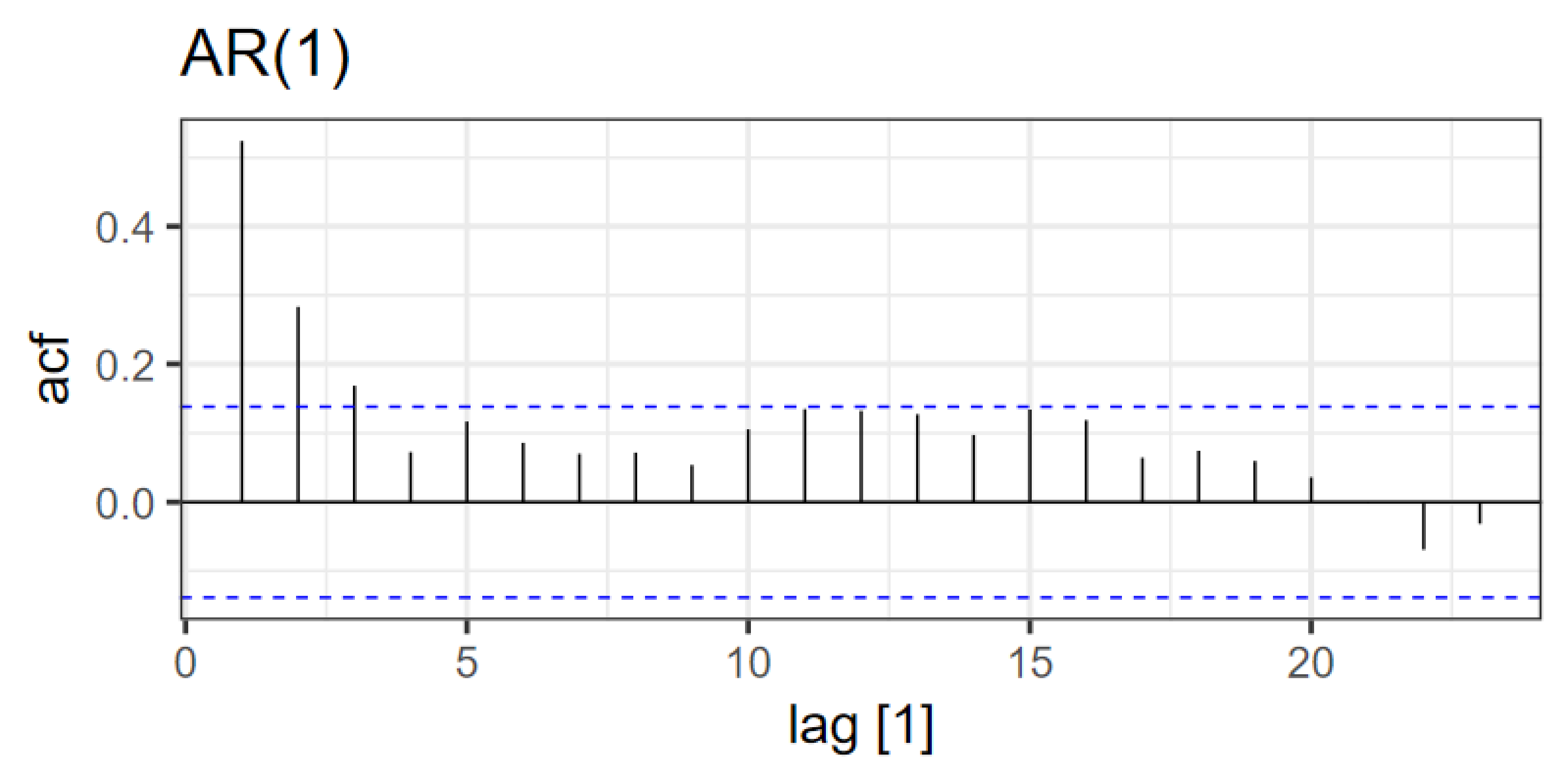
```
download.file("https://ees5891.jgilligan.org/files/data/ar_1.rds")
download.file("https://ees5891.jgilligan.org/files/data/ma_1.rds")
```

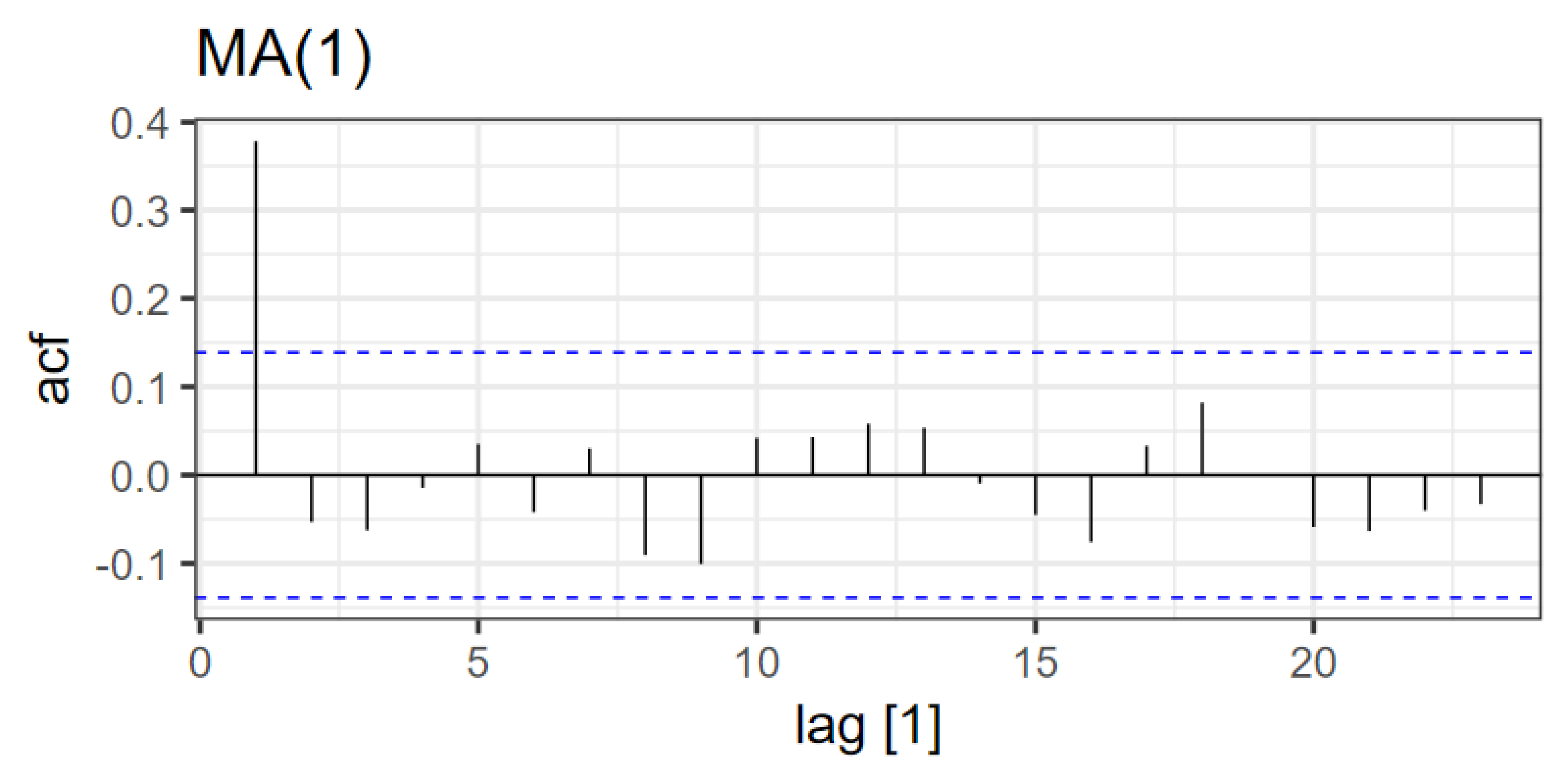
ACF

- ACF: autocorrelation function
  - For each lag k, calculate the autocorrelation of z_t with z_{t-k}
  - Plot the autocorrelation:

```
ar_1 <- read_rds("ar_1.rds")
ar_1 |> filter(phi == 0.5) |> select(t, z) |>
  as_tsibble(index = t) |> ACF() |> autoplot() +
  labs(title = "AR(1)")
ma_1 <- read_rds("ma_1.rds")
ma_1 |> filter(theta == 0.5) |> select(t, z) |>
  as_tsibble(index = t) |> ACF() |> autoplot() +
  labs(title = "MA(1)")
```

  - The blue dashed lines show the uncertainties.
    - ACF outside the lines is statistically significant
    - For MA(q), only the first q lags are significant.

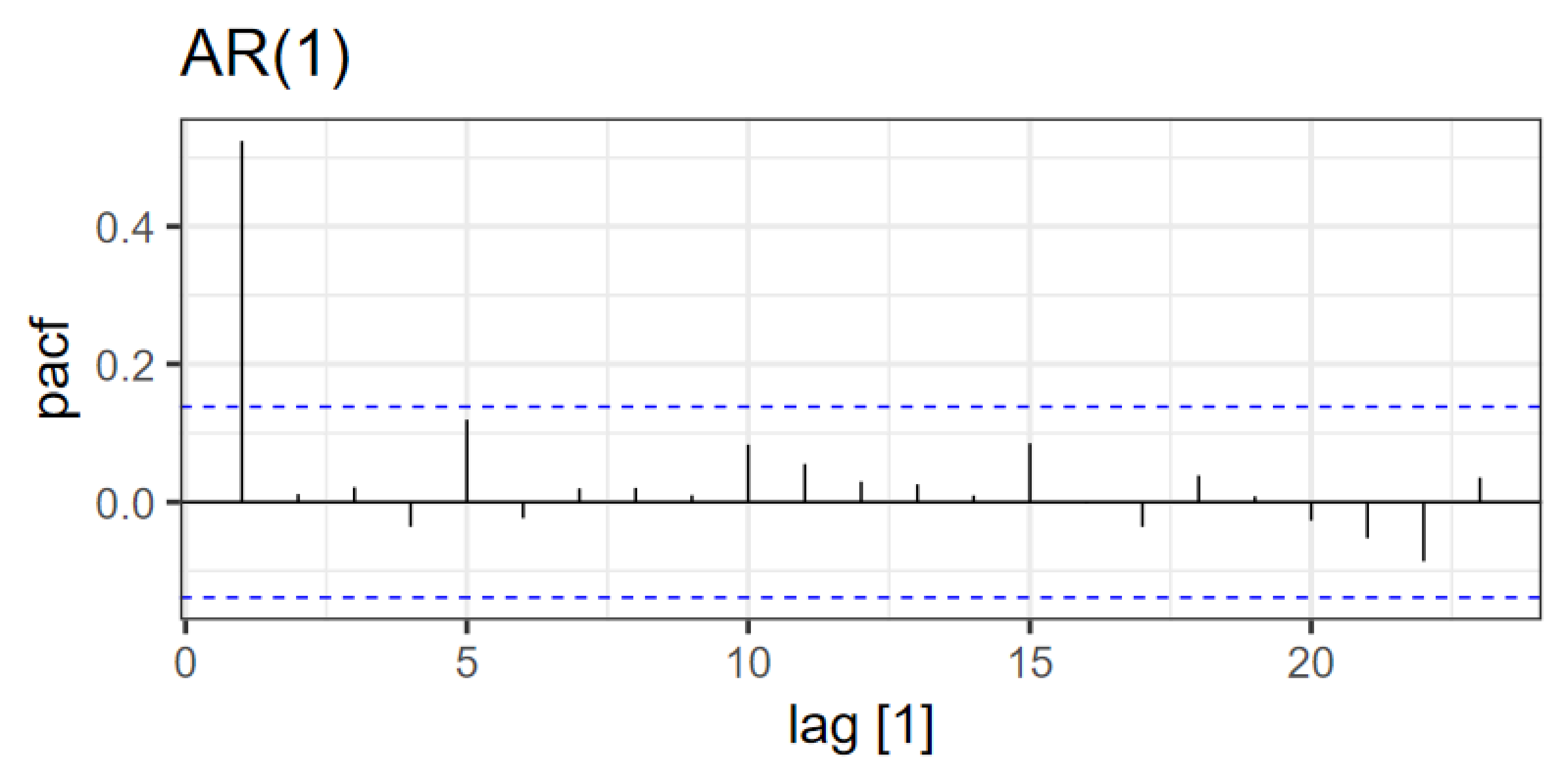


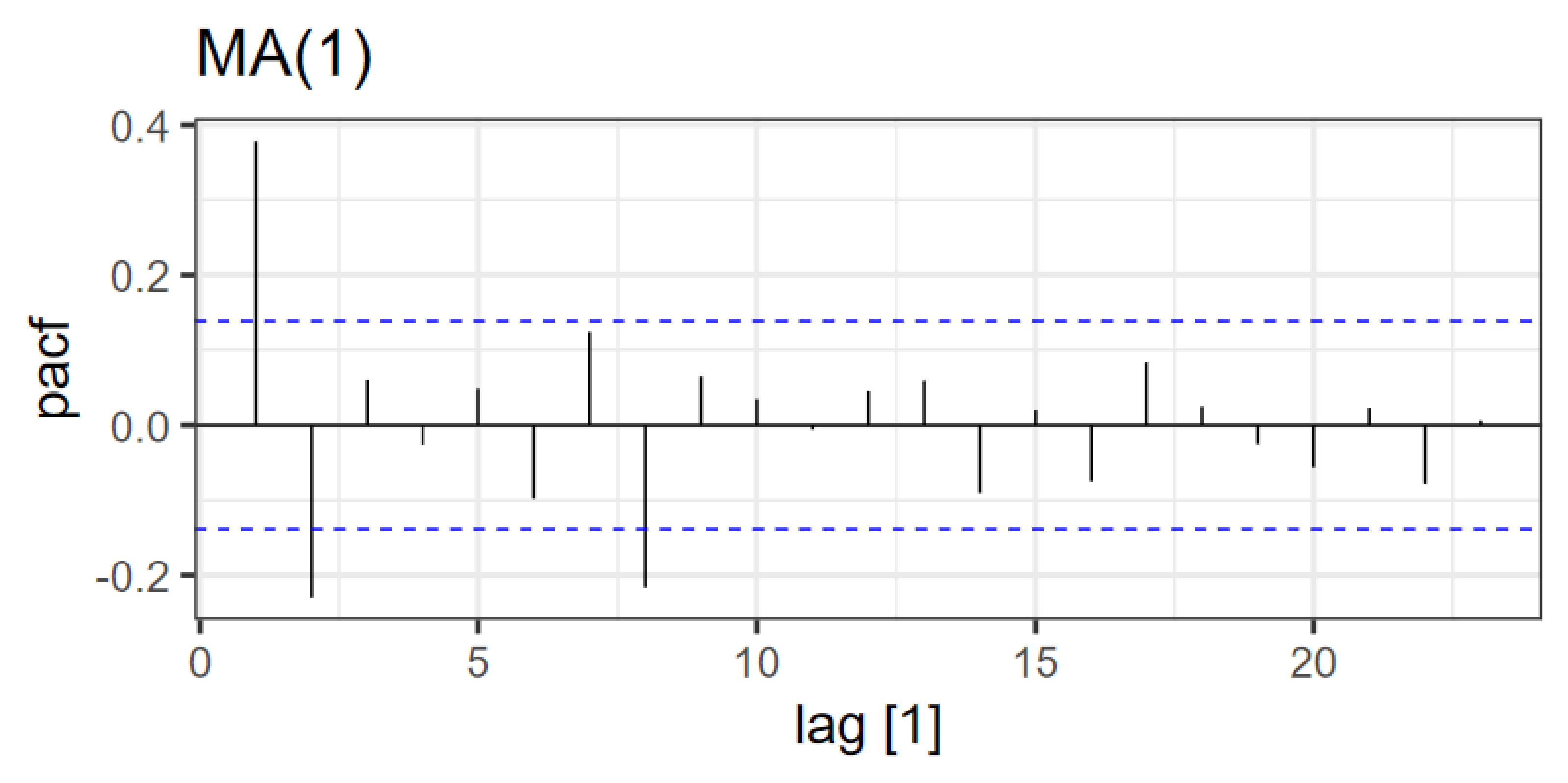


PACF

- Partial autocorrelation function
  - For each lag, calculate the autocorrelation after subtracting the autocorrelation from the previous lags
  - For AR(p), only the first p lags are significant.

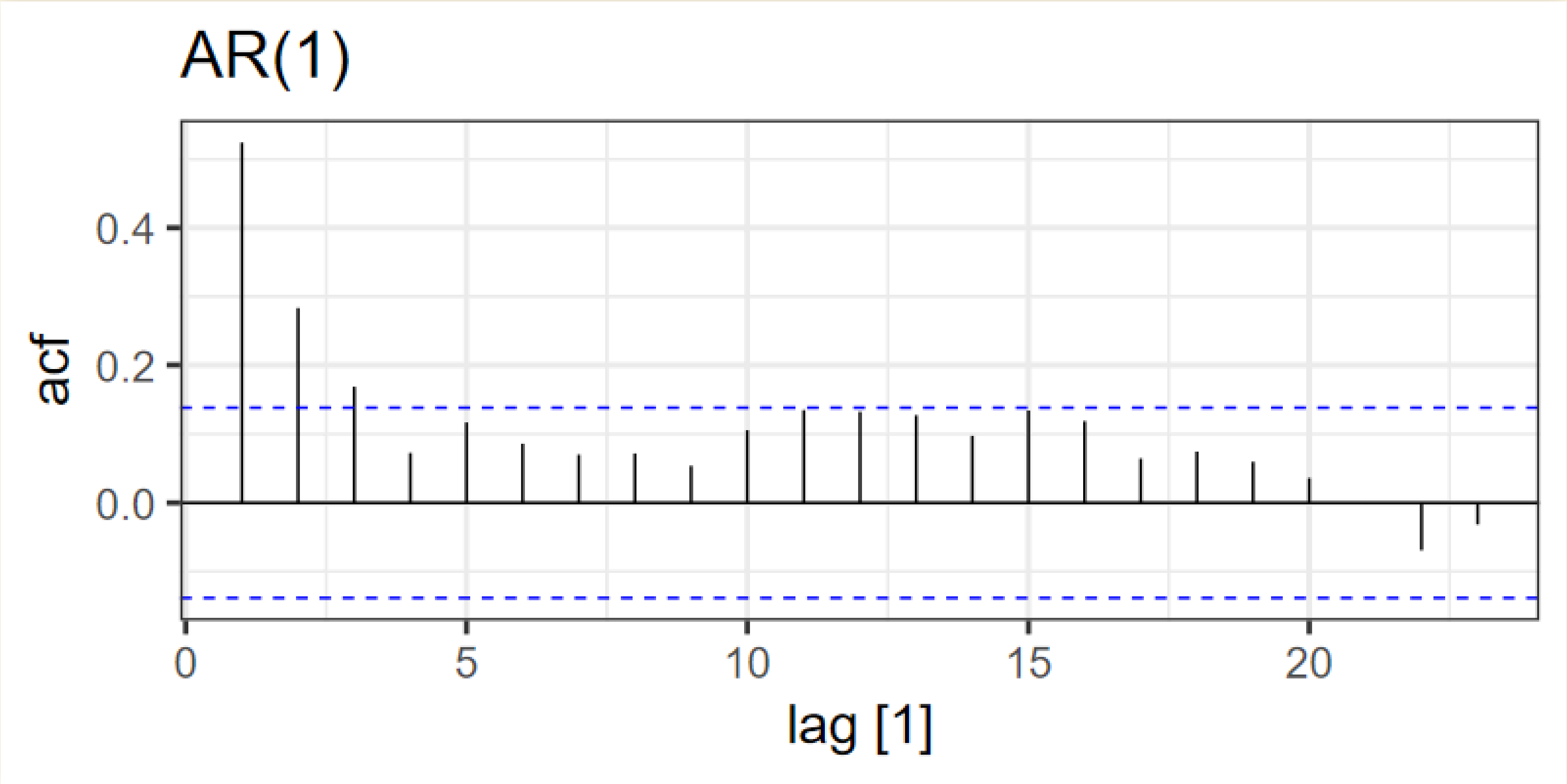
```
ar_1 |> filter(phi == 0.5) |> select(t, z) |>
  as_tsibble(index = t) |>
  PACF() |>
  autoplot() +
  labs(title = "AR(1)")
ma_1 |> filter(theta == 0.5) |> select(t, z) |>
  as_tsibble(index = t) |>
  PACF() |>
  autoplot() +
  labs(title = "MA(1)")
```

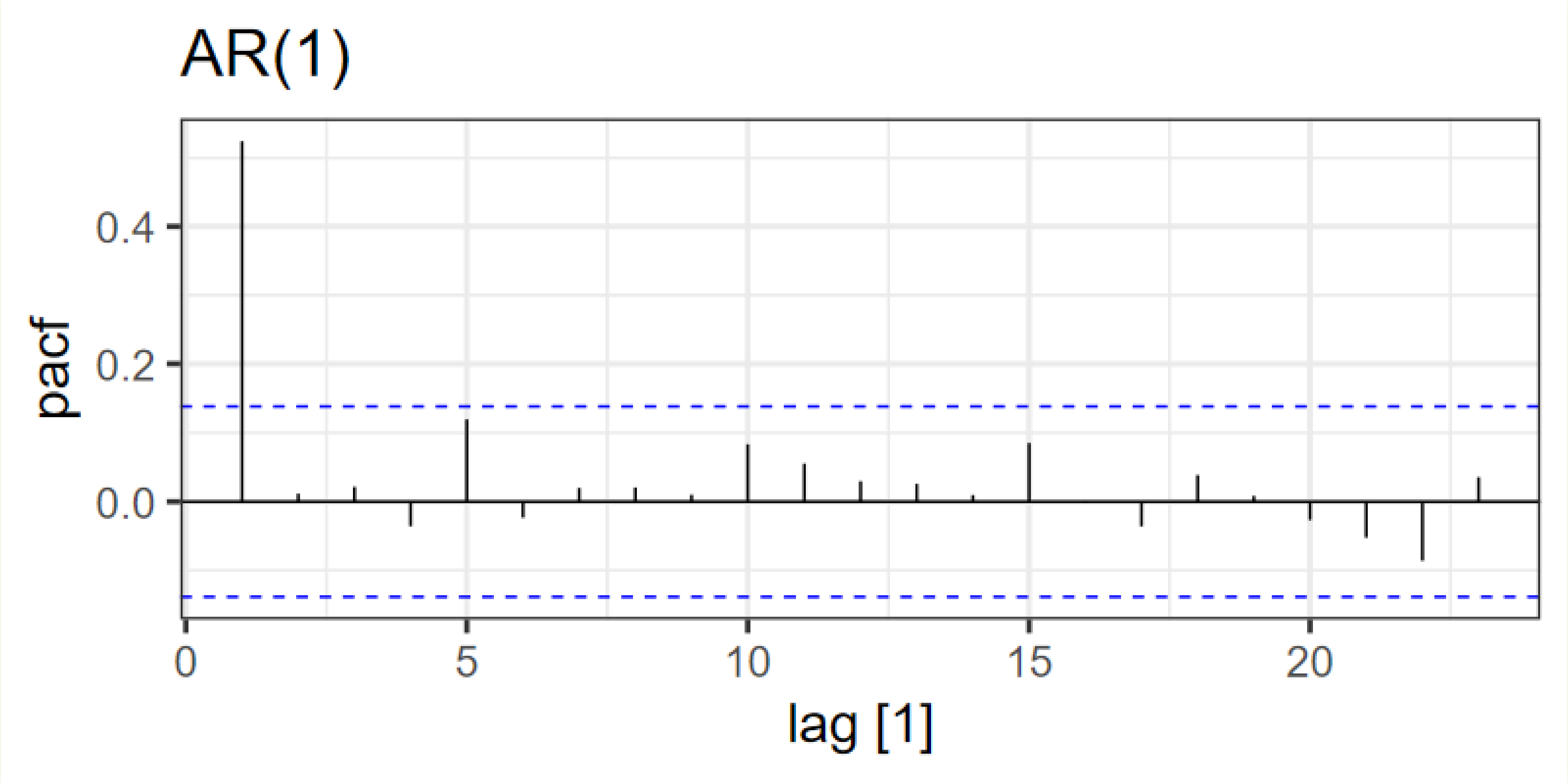




Box-Jenkins: Examine ACF and PACF

- Signature of AR(1):





- Signature of AR(1):